

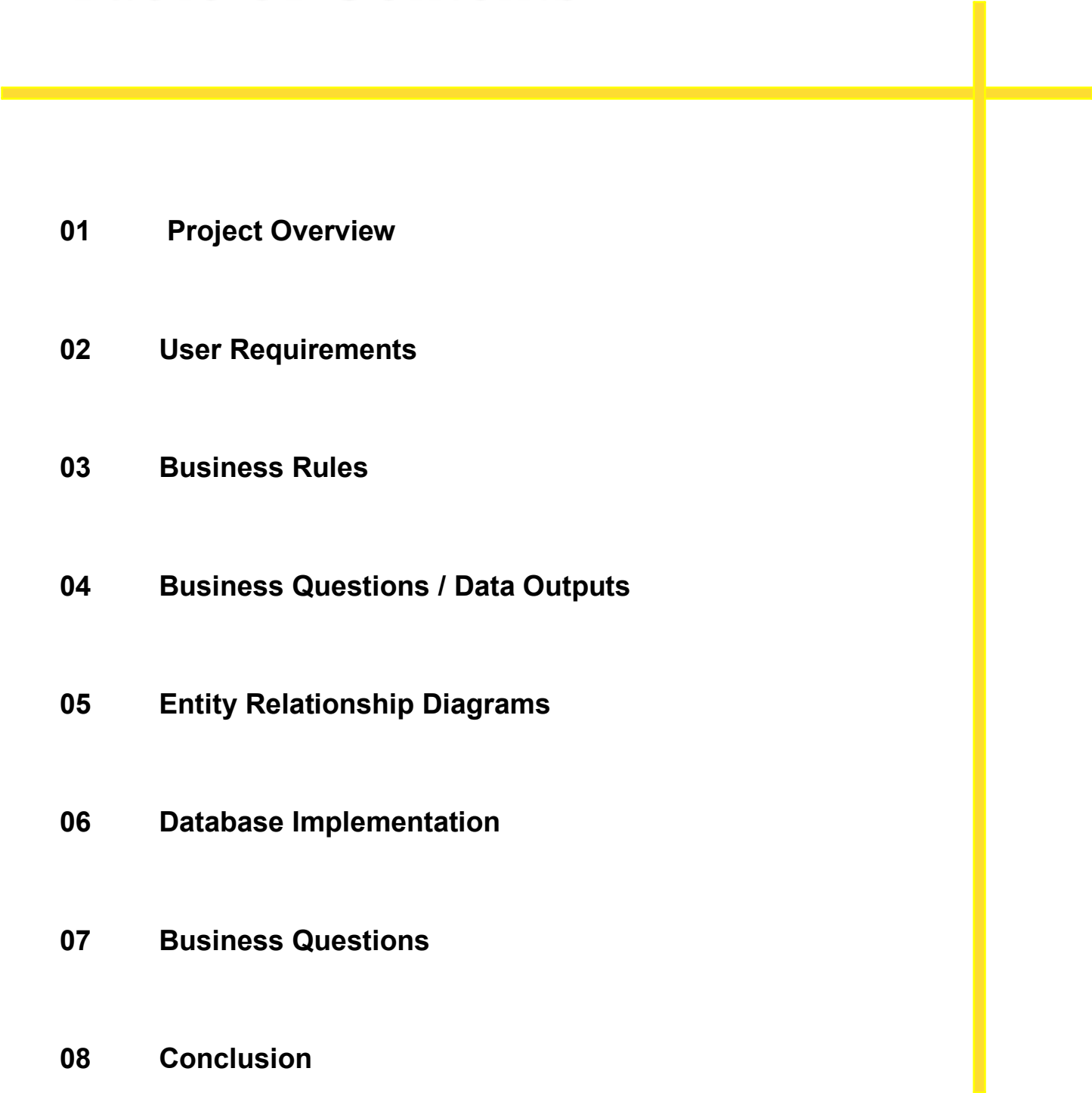


IS 4420 Final Project

IS 4420 – Spring 2025

Daxton Hunter

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Project Overview

Our group has designed a database for Expedia, specifically the airline travel section, to aid them in achieving their business requirements. Expedia is a service that earns profit from being a travel agency marketplace, where customers are given a wide array of options for travel and accommodation for the service fees that Expedia charges when booking through their website. Expedia's goals are to maximize profits for their shareholders by providing a comprehensive travel marketplace for anyone. Expedia sells flights, hotels, travel insurance, cruises and many other travel experiences.

While this project is going to focus on the biggest product of Expedia's business, the flight transaction aspect of the business. This database provides a proficient location to store all data being tracked by Expedia that includes customer, flight, and other transactional data. Using the database to ploy marketing campaigns, understand their customer base better, tracking necessary data for legal reasons, and can scale exponentially.

User Requirements (Features)

Expedia Database User Requirements are defined by the business analysis that we have provided. We have seen and stated the requirements that are needed for Expedia's database to function properly.

Data Storage Requirements

- The system must store customer personal information including first name, last name, email, and phone number for customer identification and communication/marketing purposes.
- The system must maintain flight information including flight numbers, departure/arrival times, and status to provide accurate booking options to customers.
- The system must store airport information including codes, names, cities, and time zones to support flight searches and booking operations.
- The system must record airline company details including names and headquarters locations to appropriately attribute flights to their operating airlines.
- The system must maintain aircraft information including tail numbers, models, and capacity to properly assign aircraft to scheduled flights.

Data Update Requirements

- The system must update flight status information in real-time to keep customers informed of changes.
- The system must update seat availability immediately after bookings are confirmed to prevent double-booking of seats.
- The system must modify loyalty point balances when points are earned from new bookings or redeemed for travel rewards.

- The system must update payment status information when transactions are processed, failed, or refunded.

resolution details when issues are addressed by support staff.

Data Deletion Requirements

- The system must allow for the cancellation of flight bookings while maintaining a transaction history for record-keeping and potential refund processing.
- The system must provide a secure method for customers to delete saved payment methods from their profiles while preserving transaction records.

Data Retrieval Requirements

- The system must allow customers to search available flights based on departure/arrival airports, dates, and times to find suitable travel options.
- The system must retrieve seat availability and pricing information for specific flights to present booking options to customers.
- The system must access customer loyalty information including program enrollment status and available points to apply appropriate benefits during the booking process.
- The system must retrieve baggage policy information associated with specific flights and airlines to inform customers during booking.

- The system must be able to pull up a customer's complete booking history to allow for customer service inquiries and support ticket resolution.
- The system must process reward point redemptions against bookings and calculate the adjusted price based on points used.
- The system must support tracking of travel documents including passport information, document expiration dates, and country of issue to facilitate international travel bookings.
- The system must allow customers to manage their carry-on and checked baggage preferences and track baggage information associated with specific flights.
- The system must support hotel bookings in addition to flights, linking them to the same customer profile and loyalty program.
- The system must enable customer support representatives to create, update, and resolve support tickets associated with specific customer accounts and bookings.
- The system must calculate and apply appropriate loyalty program benefits based on a customer's loyalty status and the specific flight being booked.
- The system must track aircraft maintenance status to ensure only properly maintained aircraft are assigned to scheduled flights.
- The system must maintain historical transaction records for auditing, customer service, and financial reporting purposes.

Business Rules

With these specified business rules that are derived from the entities of our physical model to ensure Expedia's database requirements are met with these guideline rules.

- Each CUSTOMER makes zero to many FLIGHT BOOKINGS, while each FLIGHT BOOKING is made by one and only one CUSTOMER.
- Each CUSTOMER can enroll in zero to one LOYALTY PROGRAM through the CUSTOMER LOYALTY PROGRAM relationship, while each LOYALTY PROGRAM can have zero to many CUSTOMERS enrolled.
- Each CUSTOMER has zero to one LOYALTY STATUS within a LOYALTY PROGRAM, while each LOYALTY STATUS can be held by zero to many CUSTOMERS.
- Each FLIGHT is operated by one and only one AIRLINE, while each AIRLINE operates zero to many FLIGHTS.
- Each FLIGHT departs from one and only one AIRPORT and arrives at one and only one AIRPORT, while each AIRPORT can be the departure point for zero to many FLIGHTS and can be the arrival point for zero to many FLIGHTS.
- Each FLIGHT is operated with one and only one AIRCRAFT, while each AIRCRAFT can be used for zero to many FLIGHTS.
- Each AIRLINE owns or leases zero to many AIRCRAFT, while each AIRCRAFT is owned or leased by one and only one AIRLINE.
- Each FLIGHT has zero to many SEATS, while each SEAT belongs to one and only one FLIGHT.
- Each FLIGHT BOOKING reserves zero to many SEATS through the BOOKING SEAT relationship, while each SEAT can be reserved through zero to one BOOKING SEAT at a time.
- Each BOOKING SEAT specifies one and only one SEAT CLASS for each reservation, while each SEAT CLASS can be assigned to zero to many BOOKING SEATS.

- Each FLIGHT BOOKING has zero to many PAYMENTS, while each PAYMENT is made for one and only one FLIGHT BOOKING.
- Each PAYMENT has zero to many entries in the TRANSACTION HISTORY, while each TRANSACTION HISTORY entry is associated with one and only one PAYMENT.
- Each CUSTOMER can make zero to many REWARD REDEMPTIONS, while each REWARD REDEMPTION is made by one and only one CUSTOMER.
- Each FLIGHT BOOKING can have zero to one REWARD REDEMPTION applied to it, while each REWARD REDEMPTION is applied to one and only one FLIGHT BOOKING.
- Each CUSTOMER can check zero to many BAGGAGE items, while each BAGGAGE item is checked by one and only one CUSTOMER for one and only one FLIGHT.
- Each FLIGHT can have one and only one BAGGAGE policy, while each BAGGAGE policy applies to one and only one FLIGHT.
- Each CUSTOMER can bring zero to many CARRY-ON items, while each CARRY-ON item belongs to one and only one CUSTOMER and is brought onto one and only one FLIGHT.
- Each CUSTOMER can have zero to many TRANSACTION HISTORY records, while each TRANSACTION HISTORY record is associated with one and only one CUSTOMER.
- Each CUSTOMER can make zero to many HOTEL BOOKINGS, while each HOTEL BOOKING is made by one and only one CUSTOMER.
- Each HOTEL BOOKING can be associated with zero to one LOYALTY PROGRAM for points accrual, while each LOYALTY PROGRAM can be linked to zero to many HOTEL BOOKINGS.
- Each CUSTOMER can have zero to many TRAVEL DOCUMENTS, while each TRAVEL DOCUMENT belongs to one and only one CUSTOMER.

- Each CUSTOMER can submit zero to many CUSTOMER SUPPORT tickets, while each CUSTOMER SUPPORT ticket is submitted by one and only one CUSTOMER.
- Each SEAT can have specific pricing in the SEATING PRICES table based on the CUSTOMER LOYALTY status and specific BOOKING details.
- Each SEATING PRICES record is associated with one and only one SEAT, one and only one CUSTOMER LOYALTY record, and one and only one BOOKING.
- Each FLIGHT BOOKING has one and only one TICKET ID, while each TICKET ID belongs to one and only one FLIGHT BOOKING.
- Each FLIGHT BOOKING has one and only one PAYMENT STATUS, which indicates whether the booking has been paid for, is pending payment, or has been refunded.
- Each BOOKING has one and only one TOTAL PRICE, which represents the final amount charged for all services included in the booking.
- Each CUSTOMER LOYALTY PROGRAM record links one and only one CUSTOMER with one and only one LOYALTY PROGRAM, creating a unique enrollment record for each customer-program combination.

Business Questions / Data output

I need a monthly report of total flight bookings and revenues, so I can adjust future pricing and promotions.

- Stakeholder: Revenue Manager
- Operational or Strategic: Strategic
- Why important: Understand peak flight booking periods to optimize pricing strategies.
- How it helps: Helps adjust seasonal pricing and maximize revenue.

- I need to analyze how many hotel bookings are made by customers enrolled in the Loyalty Program.
- Stakeholder: Marketing Director
- Operational or Strategic: Strategic
- Why important: Evaluate Loyalty Program impact on repeat business.
- How it helps: Helps refine loyalty rewards.

- I need a list of top-performing flight routes by total revenue earned.
- Stakeholder: Network Planner
- Operational or Strategic: Strategic
- Why important: Identify which flight routes generate the most revenue/
- How it helps: Helps plan future route extensions.

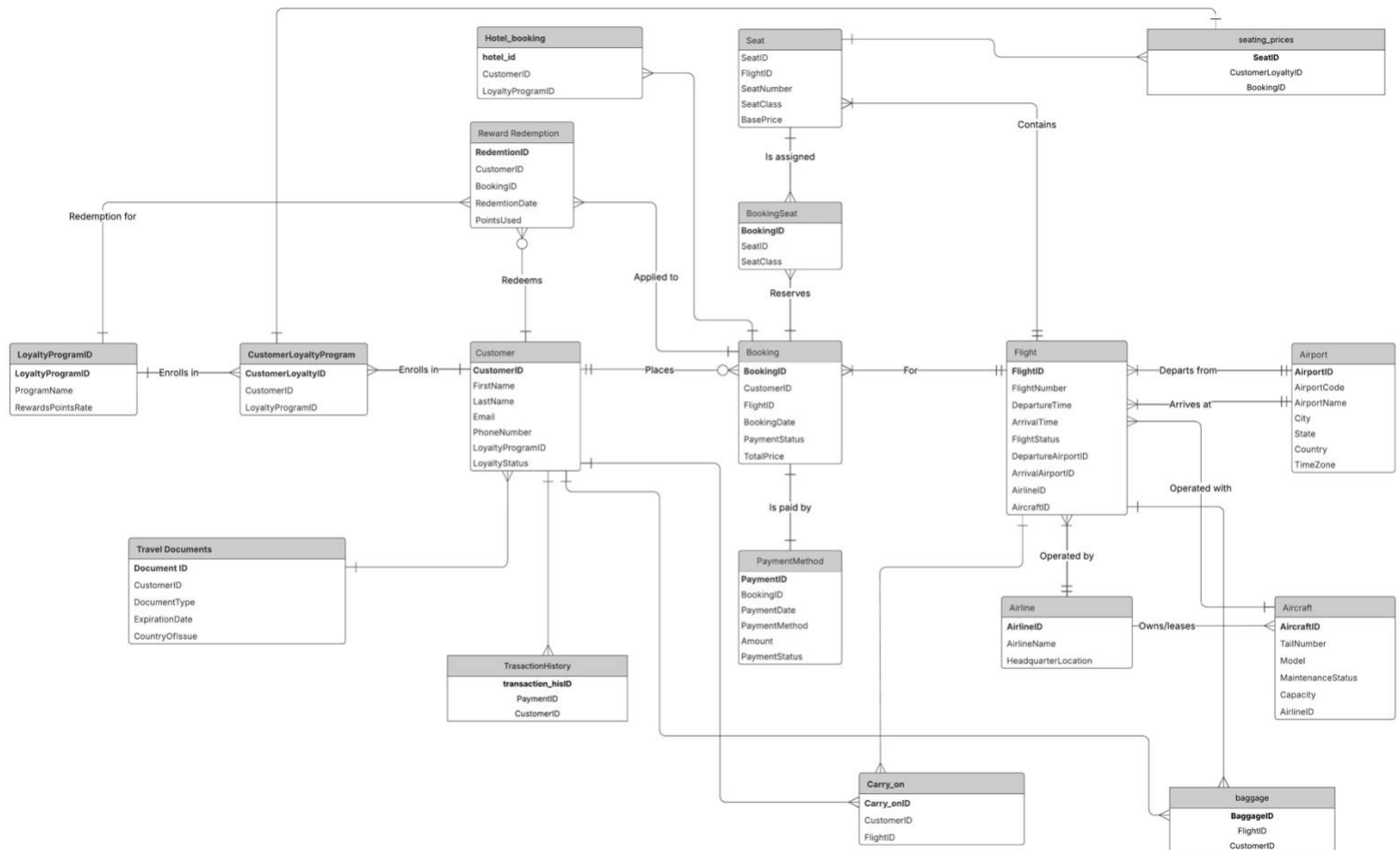
- I need a report on the count of customer support issues raised by customers by month.
- Stakeholder: Customer Service Manager
- Operational or Strategic: Operational
- Why important: Identify reoccurring issues raised by customers.
- How it helps: Improves response processes and training.

- I need a report showing aircraft types with the highest number of seat bookings.
- Stakeholder: Operations Manager
- Operational or Strategic: Strategic
- Why important: Understand which aircraft types are most booked.
- How it helps: Guides fleet planning and upgrades.
- I need a list of customers with the highest number of flights booked over the past year.
- Stakeholder: Loyalty Program Manager
- Operational or Strategic: Strategic
- Why important: Target high value customers for exclusive offers.
- How it helps: Boosts retention of top-tier customers.
- I need to analyze the average sale price per seat class (Economy, Business, First).
- Stakeholder: Pricing Manager
- Operational or Strategic: Operational
- Why important: Evaluate current seat pricing strategies across classes.
- How it helps: Fine tune pricing models to increase profitability.
- I need to see the total number of checked baggage items per month.
- Stakeholder: Finance Director
- Operational or Strategic: Operational
- Why important: Track and forecast baggage volume.
- How it helps: Ensures baggage fees meet financial targets.
- I need a list of top hotels by number of bookings made through our platform.
- Stakeholder: Strategic Partnerships Manager
- Operational or Strategic: Strategic
- Why important: Identify which hotel partners drive the most bookings.
- How it helps: Inform partnership renewals and incentives.
- I need a report of all flight bookings with a 'Pending' payment status to ensure timely collections.

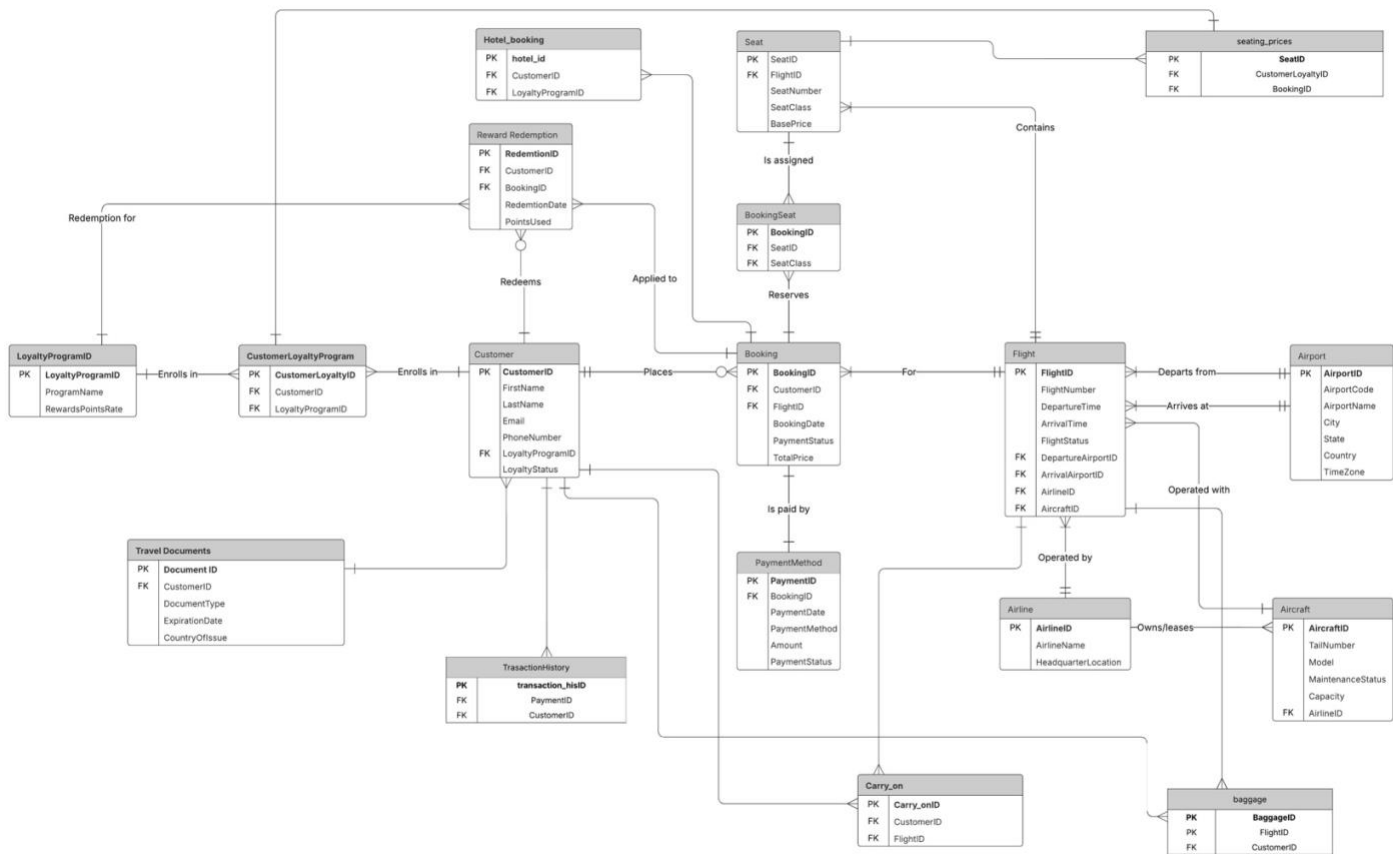
- Stakeholder: Accounting Manager
- Operational or Strategic: Operational
- Why important: Track incomplete or pending customer payments to minimize revenues leaks.
- How it helps: Helps the finance team follow up on unpaid bookings and maintain cash flow.

Entity Relationship Diagrams:

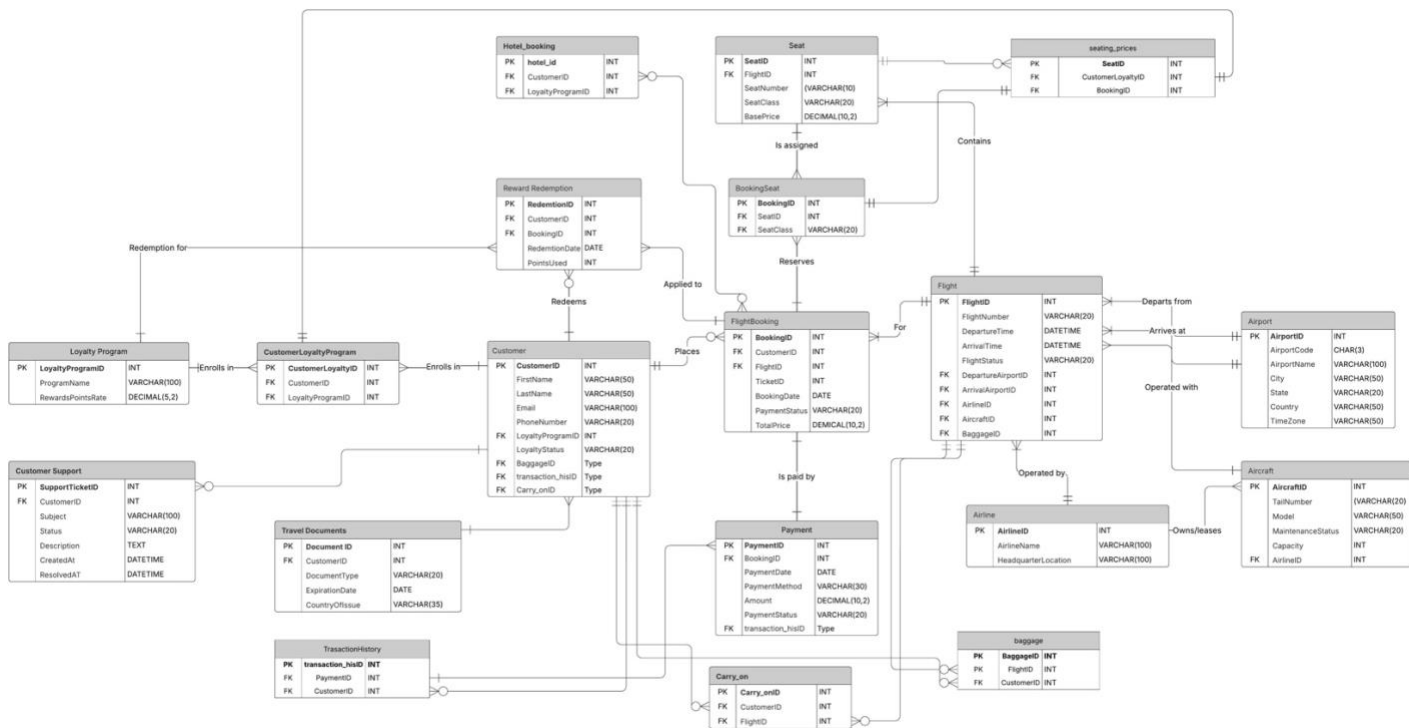
Conceptual:



Relational:



Physical:



Database Implementation

Refer to the attached script to view the entirety of the tables/script.

Table: Customer

CustomerID	FirstName	LastName	Email	PhoneNumber	LoyaltyProgramID	LoyaltyStatus	transaction_hist...	BaggageID	Carry_onID
1	Cassie	Sullivan	claytonbates@gutierrez-berry.com	058.279.0400x6095	40	Silver	12	7	18
2	Katrina	Foster	chaselam@swanson.com	265.050.5727x35594	36	Silver	14	10	18
3	Omar	Williams	zolson@swanson.info	(197)926-3950x33075	30	Gold	31	33	17
4	David	Sullivan	changmeghan@chavez.com	558-165-6792x03340	2	Platinum	20	11	26
5	Michelle	George	morganmatthew@washington.info	341-508-0775x651	17	Platinum	37	33	44
6	Erica	Russell	kellyanderson@fischer.com	(701)765-9068x37875	5	Platinum	17	35	43
7	Todd	Vincent	josephoconnell@gmail.com	(304)180-6411x758	6	Silver	15	35	45
8	Ashley	Medina	andersonrussell@hotmail.com	+1-310-775-9525x6984	25	Platinum	40	12	2
9	Joshua	Campbell	brittanygilbert@jones-hoffman.com	186-354-7785x846	31	Platinum	9	2	39
10	Lucas	Gonzalez	smithsherry@hotmail.com	001-893-127-5937	32	Gold	5	30	1
11	Dalton	Mendoza	wilcoxcrraig@gmail.com	001-586-462-6190x115	39	Platinum	15	42	24
12	Paula	Cisneros	mfisher@yahoo.com	282-176-1980	25	Silver	1	38	45
13	Diana	Payne	martinbrian@young.com	(483)685-7337x87343	23	Gold	18	42	31
14	Joel	Jordan	gordonaaron@gmail.com	358.863.5833x851	9	Platinum	1	22	29
15	Carl	Buck	kristenguzman@gmail.com	531-622-8862x0244	16	Platinum	3	37	11
16	Angela	Williams	lauren01@wright-chavez.com	(705)636-9702	41	Gold	44	25	34
17	Christop...	Williams	griffinmatthew@gmail.com	505-509-5681	38	Platinum	12	26	23
18	Stephen	Roberts	hamptonmelissa@hotmail.com	(803)377-9707x03309	37	Gold	14	19	22
19	Todd	Lamb	john05@valdez.biz	+1-886-096-8135x6600	14	Platinum	44	28	2
20	Alejandro	Jefferson	emilyharvey@thompson-swanso...	001-399-220-8737x629	26	Silver	1	38	39

Table: Airport

AirportID	AirportCo...	AirportName	City	State	Country	TimeZone
1	OES	Port Catherine Airport	East Vickie	Colorado	Mexico	America/El_Salvador
2	YKY	Port Olivia Airport	Justintown	Utah	Canada	Asia/Vladivostok
3	CJH	Henrysides Airport	West Gregorymouth	Oklahoma	United States	Europe/Andorra
4	LEO	Kentfurt Airport	North Matthewbury	Florida	United States	Pacific/Fiji
5	USU	Hilton Airport	Katherinehaven	Missouri	Germany	Asia/Kuching
6	VJG	Lake Lisa Airport	West Debramouth	West Virginia	United States	Africa/Bamako
7	ZVY	Katrinafort Airport	North Cory	Washington	Germany	Africa/Asmera
8	UJK	Port Holly Airport	South Brian	North Dakota	United States	Africa/Ceuta
9	RDV	Manuelberg Airport	South Jamesside	Maine	Mexico	Pacific/Fiji
10	RWD	New Sarah Airport	Gainesland	South Dakota	Canada	Asia/Kabul
11	KID	Dyerside Airport	Douglaston	Colorado	Mexico	Europe/Podgorica
12	WAQ	New Juliaburgh Airport	New Luke	Wyoming	United States	Asia/Yerevan
13	GQU	Harrisonfort Airport	Christyfort	Texas	Canada	Africa/Gaborone
14	YHX	Meganburgh Airport	Colemouthe	Maryland	United States	Asia/Beirut
15	CBP	Birdport Airport	West Ericmouth	Wyoming	Mexico	America/Guatemala
16	JHU	Charleschester Airport	Jeanton	Alabama	Germany	Asia/Aden
17	VMP	Johnnyland Airport	West Todd	Michigan	United States	Asia/Riyadh
18	CJX	Kellystad Airport	Rodriguezside	Georgia	Germany	Africa/Maputo
19	TYC	New Edward Airport	Port Sarahon	Missouri	United States	Asia/Vientiane
20	RHE	Lopezfurt Airport	Chelseaburgh	Maine	Germany	America/Belize

Table: Airline

	AirlineID	AirlineName	HeadquarterLocati...	
1		SkyJet Airlines	Shepherdstad	
2		GlobeAir Airlines	North Jennifermouth	
3		StarLink Airlines	Lake Justin	
4		AeroNova Airlines	North Hollyfort	
5		SkyJet Airlines	Michaelfort	
6		GlobeAir Airlines	Wolfeview	
7		StarLink Airlines	Port Vincentstad	
8		AeroNova Airlines	Melissaberg	
9		SkyJet Airlines	Loriside	
10		GlobeAir Airlines	West Carrietown	
11		StarLink Airlines	Johnsonport	
12		AeroNova Airlines	Kingborough	
13		SkyJet Airlines	Lake Jeanne	
14		GlobeAir Airlines	Port Matthew	
15		StarLink Airlines	Port Jerry	
16		AeroNova Airlines	Blakebury	
17		SkyJet Airlines	Jeanetteborough	
18		GlobeAir Airlines	Andersonmouth	
19		StarLink Airlines	Mackland	
20		AeroNova Airlines	East Autumnstad	

Table: Aircraft

	AircraftID	TailNumber	Model	MaintenanceStat...	Capacity	AirlineID	
1		T-6701	Before Jet	Operational	192	37	
2		T-6873	Peace Jet	Operational	160	5	
3		T-7901	With Jet	Operational	188	27	
4		T-2065	Agent Jet	Operational	212	35	
5		T-3084	After Jet	Maintenance	147	8	
6		T-9603	Music Jet	Maintenance	299	36	
7		T-3322	Cell Jet	Maintenance	198	24	
8		T-5832	Million Jet	Operational	338	21	
9		T-2355	Try Jet	Operational	295	37	
10		T-4701	Move Jet	Maintenance	262	14	
11		T-4908	Over Jet	Operational	256	14	
12		T-3629	Stop Jet	Maintenance	216	12	
13		T-9266	Strategy...	Operational	199	12	
14		T-5342	Evening...	Maintenance	224	15	
15		T-1893	Side Jet	Maintenance	324	15	
16		T-4828	Congres...	Operational	324	2	
17		T-2024	Born Jet	Operational	140	15	
18		T-8301	Official Jet	Maintenance	126	44	
19		T-6216	Produce...	Maintenance	335	12	
20		T-8341	Leader Jet	Operational	178	20	

Table: Flight

	FlightID	FlightNumber	DepartureTime	ArrivalTime	FlightStatus	DepartureAirport...	ArrivalAirport...	AirlineID	AircraftID	
1		FL8375	2024-08-16 18:26:48	2025-05-24 15:11:38	Delayed	22	10	7	37	
2		FL6978	2024-07-04 05:24:20	2025-04-26 23:51:56	On Time	45	44	4	45	
3		FL5797	2024-10-27 02:31:30	2025-05-15 10:42:23	On Time	12	18	28	16	
4		FL0415	2025-03-16 00:33:07	2025-04-30 18:49:21	On Time	44	21	21	24	
5		FL9174	2025-02-07 03:04:45	2025-05-12 17:11:17	Delayed	7	23	26	11	
6		FL4438	2024-06-14 18:27:03	2025-05-22 18:26:16	Delayed	23	29	31	41	
7		FL9039	2024-12-31 06:57:03	2025-05-06 17:01:52	On Time	2	20	38	29	
8		FL1679	2025-02-09 21:56:36	2025-05-19 17:33:06	On Time	33	1	37	25	
9		FL5822	2024-05-03 20:25:42	2025-05-10 21:58:41	Delayed	9	43	34	42	
10		FL8766	2024-04-30 06:24:55	2025-05-16 01:54:31	On Time	7	5	2	25	
11		FL3335	2024-05-18 02:11:03	2025-05-13 20:58:55	On Time	3	31	31	3	
12		FL1358	2025-04-14 07:48:53	2025-05-19 01:28:37	Delayed	18	28	31	35	
13		FL9812	2024-05-06 11:07:13	2025-05-15 18:02:26	Delayed	28	36	8	27	
14		FL7749	2025-02-01 03:47:09	2025-05-23 14:22:59	Delayed	13	31	9	11	
15		FL4357	2024-12-09 12:08:32	2025-05-21 05:25:49	On Time	39	11	30	42	
16		FL2389	2024-09-28 03:20:06	2025-04-28 19:02:12	Delayed	24	37	32	23	
17		FL3735	2025-04-16 08:24:16	2025-05-13 15:53:00	On Time	24	42	43	7	
18		FL5096	2024-08-10 12:30:51	2025-05-15 23:12:44	Delayed	12	28	39	22	
19		FL0565	2025-02-25 10:19:12	2025-05-11 08:03:19	On Time	38	11	40	42	
20		FL5214	2025-02-14 02:59:38	2025-05-12 09:22:26	Delayed	30	27	18	4	

Table: Loyalty Program

LoyaltyProgramID	ProgramName	RewardsPointsRate	
1	Silver Circle	2.04	
2	Gold Elite	1.95	
3	Platinum Pro	3.07	
4	Traveler Club	2.28	
5	Silver Circle	2.59	
6	Gold Elite	3.03	
7	Platinum Pro	1.39	
8	Traveler Club	1.53	
9	Silver Circle	1.53	
10	Gold Elite	2.55	
11	Platinum Pro	1.07	
12	Traveler Club	2.36	
13	Silver Circle	3.11	
14	Gold Elite	3.43	
15	Platinum Pro	1.79	
16	Traveler Club	1.03	
17	Silver Circle	2.34	
18	Gold Elite	3.28	
19	Platinum Pro	3.02	
20	Traveler Club	3.61	

Table: Reward Redemption

RedemptionID	CustomerID	BookingID	RedemptionDate	PointsUsed	
1	23	1	2023-01-11	866	
2	34	2	2023-04-10	8012	
3	42	3	2022-09-23	7432	
4	40	4	2022-08-28	6089	
5	12	5	2025-12-06	9123	
6	38	6	2023-02-08	7514	
7	4	7	2022-09-07	8608	
8	13	8	2024-12-24	7869	
9	18	9	2023-02-09	4461	
10	44	10	2022-10-02	4412	
11	2	11	2022-10-27	1315	
12	3	12	2023-03-04	5097	
13	36	13	2025-10-27	4447	
14	39	14	2022-01-20	1595	
15	28	15	2025-03-12	5433	
16	33	16	2024-09-10	3272	
17	8	17	2025-02-12	8329	
18	1	18	2025-03-28	2608	
19	5	19	2022-06-22	1138	
20	28	20	2024-11-15	8905	

Table: Customer Support

SupportTicket...	CustomerID	Subject	Status	Description	CreatedAt	ResolvedAt	
1	12	Argue national prepare.	Open	Three too project agent war public. See simple i...	2024-05-10 10:56:01	2024-05-23 10:56:01	
2	20	Watch make name.	Open	Fire defense buy open over few not. Night from...	2024-12-01 17:32:52	2024-12-25 17:32:52	
3	6	Well support laugh.	Open	Another result support. Some new benefit like.	2023-07-16 17:12:30	2023-08-09 17:12:30	
4	8	Another popular girl body suddenly.	Open	Administration politics less the race economy hi...	2024-08-12 18:44:06	2024-09-09 18:44:06	
5	24	Else free probably.	Closed	Unit whose many enter town rate contain remai...	2023-06-29 05:45:07	2023-07-15 05:45:07	
6	30	Get protect hit at.	Closed	Song happy include get total. Visit a cold medic...	2023-11-12 23:07:48	2023-11-16 23:07:48	
7	35	Focus how few.	Open	Wide action size top newspaper. Evidence even...	2023-10-12 08:43:38	2023-10-13 08:43:38	
8	45	Recently foreign never.	Closed	Nation talk TV charge scene certainly. Election c...	2024-05-31 20:50:44	2024-06-19 20:50:44	
9	34	Someone unit second floor smile.	Closed	Study actually skill. Week group both skill late pi...	2023-12-31 09:49:03	2024-01-13 09:49:03	
10	29	Mother building can director three.	Open	Pull leg thank today treatment him. Television pr...	2023-05-09 12:46:12	2023-06-05 12:46:12	
11	18	Fire generation to try.	Open	Worker dog office send. Lose will writer.	2024-07-17 04:44:31	2024-07-20 04:44:31	
12	9	Within simply.	Closed	Old city knowledge shake save type. Read orga...	2024-09-23 15:23:23	2024-09-24 15:23:23	
13	21	Same very kitchen school.	Closed	Phone nor red major create herself. Off west dir...	2023-05-15 17:00:16	2023-06-04 17:00:16	
14	43	Section both.	Open	Stay fight off son read leave. Stage young bring...	2024-04-22 14:36:25	2024-05-07 14:36:25	
15	36	Involve war cover long.	Closed	Either skin increase PM. Size camera letter. Top...	2024-01-16 10:42:47	2024-01-19 10:42:47	
16	3	Compare build grow.	Closed	Money language goal similar. Institution really e...	2023-07-30 22:15:01	2023-08-10 22:15:01	
17	10	Pass statement leg toward.	Open	Computer picture participant reduce hour news f...	2023-06-04 11:36:42	2023-06-27 11:36:42	
18	21	Dinner down bank keep.	Closed	Hundred research home true city garden. Now h...	2024-09-04 20:36:42	2024-09-28 20:36:42	
19	20	To issue different Republican attor...	Open	Cover concern development way speak commo...	2023-11-23 23:14:28	2023-11-26 23:14:28	
20	3	Prevent day.	Closed	See religious north wrong win. Start clear camp...	2025-03-02 21:46:01	2025-03-10 21:46:01	

Table: Travel Documents

	DocumentID	CustomerID	DocumentType	ExpirationDate	CountryOfIssue	
1		27	Passport	2030-04-11	Canada	
2		4	Passport	2028-07-30	Germany	
3		23	Visa	2030-10-06	United States	
4		19	Visa	2026-10-15	Japan	
5		26	Passport	2026-09-01	Mexico	
6		28	Visa	2030-10-03	Canada	
7		44	Visa	2030-05-13	United States	
8		32	Visa	2027-02-17	United States	
9		38	Visa	2028-01-14	Mexico	
10		12	Passport	2026-02-05	Canada	
11		25	Passport	2028-10-07	Germany	
12		19	Visa	2030-05-10	Mexico	
13		24	Passport	2029-06-11	Japan	
14		36	Passport	2028-02-23	Canada	
15		42	Passport	2029-06-01	Mexico	
16		31	Passport	2030-08-04	Germany	
17		29	Visa	2029-10-05	Germany	
18		26	Passport	2026-06-17	Canada	
19		9	Passport	2028-10-24	Mexico	
20		39	Passport	2026-11-23	Canada	

Table: Transaction History

transaction_hist...	PaymentID	CustomerID	
1	1	26	
2	2	7	
3	3	35	
4	4	5	
5	5	2	
6	6	31	
7	7	16	
8	8	35	
9	9	43	
10	10	30	
11	11	8	
12	12	44	
13	13	38	
14	14	45	
15	15	30	
16	16	34	
17	17	28	
18	18	22	
19	19	11	
20	20	39	

Table: Payment

PaymentID	BookingID	PaymentDate	PaymentMethod	Amount	PaymentStatus	transaction_hist...	
1	1	2022-06-21	Bank Transfer	2380.51	Pending	6	
2	2	2025-11-10	PayPal	713.62	Pending	44	
3	3	2023-03-18	Bank Transfer	2811.38	Completed	28	
4	4	2025-12-19	Bank Transfer	453.35	Pending	1	
5	5	2022-04-01	Bank Transfer	1274.89	Completed	42	
6	6	2025-07-02	PayPal	1059.02	Pending	14	
7	7	2022-03-17	PayPal	796.80	Completed	35	
8	8	2022-11-18	Credit Card	2448.97	Completed	31	
9	9	2025-04-26	Credit Card	2208.97	Completed	24	
10	10	2023-01-30	Bank Transfer	1043.37	Pending	33	
11	11	2022-04-17	Credit Card	1543.49	Completed	42	
12	12	2022-08-04	Credit Card	411.50	Completed	26	
13	13	2024-11-09	Credit Card	2039.44	Pending	43	
14	14	2024-02-03	Credit Card	2795.34	Pending	36	
15	15	2024-05-13	Credit Card	1981.25	Completed	6	
16	16	2024-08-15	PayPal	2731.79	Pending	16	
17	17	2025-08-14	Credit Card	401.93	Completed	34	
18	18	2022-07-31	Credit Card	1418.27	Completed	43	
19	19	2023-01-14	Credit Card	2542.13	Completed	9	
20	20	2025-03-23	Credit Card	1552.67	Completed	38	

Table: Booking Seat

	BookingID	SeatID	SeatClass	
	1	6	Business	
	1	23	Business	
	1	35	First	
	2	6	First	
	2	8	Economy	
	2	16	Economy	
	2	35	Business	
	4	24	First	
	5	19	First	
	9	7	First	
	9	36	Business	
	9	40	Economy	
	10	3	Business	
	11	13	First	
	11	16	Business	
	12	11	First	
	15	25	Business	
	18	26	First	
	20	10	Economy	
	20	17	Economy	

Table: Carry-on

	Carry_onID	CustomerID	FlightID	
	1	21	36	
	2	31	31	
	3	31	33	
	4	4	5	
	5	38	8	
	6	31	5	
	7	10	42	
	8	37	35	
	9	25	40	
	10	45	11	
	11	20	43	
	12	41	28	
	13	28	32	
	14	21	17	
	15	38	15	
	16	1	5	
	17	36	13	
	18	13	21	
	19	39	2	
	20	23	34	

Table: Customer Loyalty Program

	CustomerLoyalty...	CustomerID	LoyaltyProgramID	
	1	44	1	
	2	1	5	
	3	43	5	
	4	13	2	
	5	35	3	
	6	33	3	
	7	32	2	
	8	36	4	
	9	27	3	
	10	10	1	
	11	36	3	
	12	4	5	
	13	12	2	
	14	16	5	
	15	37	4	
	16	11	3	
	17	22	1	
	18	35	1	
	19	34	5	
	20	26	3	

Table: Flight Booking

BookingID	CustomerID	FlightID	TicketID	BookingDate	PaymentStatus	TotalPrice
1	12	28	1042	2025-08-26 00:00:00	Pending	868.00
2	32	15	1040	2025-08-12 00:00:00	Pending	574.00
3	25	42	1029	2025-07-25 00:00:00	Paid	412.00
4	41	25	1047	2025-07-16 00:00:00	Pending	931.00
5	32	38	1007	2025-07-23 00:00:00	Paid	293.00
6	21	41	1026	2025-07-24 00:00:00	Pending	396.00
7	5	28	1028	2025-06-10 00:00:00	Paid	1451.00
8	30	13	1017	2025-05-14 00:00:00	Pending	829.00
9	38	26	1045	2025-06-26 00:00:00	Paid	813.00
10	42	28	1042	2025-05-17 00:00:00	Pending	587.00
11	24	6	1013	2025-05-12 00:00:00	Paid	942.00
12	8	34	1050	2025-06-22 00:00:00	Paid	445.00
13	32	31	1034	2025-07-15 00:00:00	Pending	1168.00
14	2	30	1032	2025-07-26 00:00:00	Pending	1476.00
15	10	4	1040	2025-07-17 00:00:00	Pending	1026.00
16	3	42	1016	2025-08-19 00:00:00	Paid	950.00
17	26	2	1043	2025-05-17 00:00:00	Paid	1231.00
18	33	45	1005	2025-09-16 00:00:00	Paid	863.00
19	12	7	1022	2025-08-13 00:00:00	Pending	984.00
20	13	10	1042	2025-05-27 00:00:00	Paid	1496.00

Table: Hotel-Booking

Hotel_ID	CustomerID	LoyaltyProgramID
1	26	13
2	21	9
3	25	13
4	6	19
5	26	7
6	30	12
7	41	14
8	44	2
9	26	3
10	4	18
11	31	10
12	27	5
13	32	20
14	8	9
15	10	18
16	17	18
17	38	13
18	43	10
19	41	1
20	29	19

Table: Seat

SeatID	FlightID	SeatNumber	SeatClass	BasePrice
1	25	B27	First	649.00
2	15	A11	First	271.00
3	29	A17	First	706.00
4	40	D12	Economy	765.00
5	15	B8	Economy	294.00
6	23	B11	Business	725.00
7	2	D18	Business	218.00
8	12	A5	Economy	732.00
9	16	A7	First	647.00
10	20	A26	Economy	710.00
11	8	E2	Economy	430.00
12	30	C23	First	689.00
13	11	A18	Business	371.00
14	26	C13	Business	232.00
15	36	B13	Economy	765.00
16	33	B15	Economy	118.00
17	39	E9	Economy	664.00
18	29	A12	Business	342.00
19	21	E25	First	337.00
20	31	C3	Economy	329.00

Table: Seat-Pricing

SeatID	CustomerLoyalty...	BookingID	
7	505	1	
5	506	39	
16	506	14	
28	507	27	
5	508	28	
9	508	2	
10	508	15	
29	508	4	
40	508	10	
34	510	30	
28	515	26	
9	516	4	
22	516	32	
31	516	39	
38	516	13	
6	517	19	
11	520	2	
8	524	3	
21	524	31	
36	524	36	

Table: Baggage

BaggageID	FlightID	CustomerID	
1	24	4	
2	3	22	
3	31	33	
4	16	34	
5	7	1	
6	21	14	
7	7	41	
8	38	42	
9	5	29	
10	40	32	
11	29	28	
12	23	38	
13	27	33	
14	6	11	
15	35	17	
16	42	31	
17	18	17	
18	39	12	
19	2	16	
20	10	36	

Business Questions

- I need a monthly report of total flight bookings and revenues, so I can adjust future pricing and promotions.


```
-- 1. I need a monthly report of total flight bookings and revenues, so I can adjust future pricing and promotions.
select
    Month(BookingDate) as Month,
    count(BookingID) as TotalBookings,
    sum(TotalPrice) as TotalRevenue
from
    FlightBooking
group by
    Month(BookingDate)
order by
    Month;
```
- Answer: July recorded the highest number of bookings at 10, but with a lower total revenue of \$6,695, implying a decrease in the average ticket price. This

could indicate promotional pricing strategies or increased competition during the summer travel season.

- I need to analyze how many hotel bookings are made by customers enrolled in the Loyalty Program.

```
-- 2. I need to analyze how many hotel bookings are made by customers enrolled in the Loyalty Program.  
select  
    hl.CustomerID,  
    count(hl.Hotel_ID) as TotalHotelBookings  
from  
    Hotel_booking hl  
join  
    CustomerLoyaltyProgram clp on hl.CustomerID = clp.CustomerID  
group by  
    hl.CustomerID;
```

-
- Answer: Loyalty program members made between 1 to 4 hotel bookings each, with a few customers booking at the highest frequency (4 bookings). This suggests that while most loyalty members only booked once or twice, targeting top repeat hotel bookers with exclusive offers could further strengthen loyalty and boost hotel revenue.

- I need a list of top-performing flight routes by total revenue earned.

```
-- 3. I need a list of top-performing flight routes by total revenue earned.  
select  
    f.DepartureAirportID,  
    f.ArrivalAirportID,  
    sum(fb.TotalPrice) as TotalRevenue  
from  
    Flight f  
join  
    FlightBooking fb on f.FlightID = fb.FlightID  
group by  
    f.DepartureAirportID, f.ArrivalAirportID  
order by  
    TotalRevenue desc;
```

-
- Answer: The route from Departure AirportID 15 to Arrival AirportID 34 generated the highest revenue (\$4,403), significantly outperforming all other

routes.

This suggests that this specific route is a major revenue driver and could be prioritized for additional capacity, targeted promotions, or dynamic pricing strategies.

- I need a report on the count of customer support issues raised by customers by month.

```
-- 4. I need a report on the count of customer support issues raised by customers by month.
select
    Month(CreatedAt) as Month,
    count(SupportTicketID) as Total_Issues
from
    CustomerSupport
group by
    Month(CreatedAt)
order by
    Month;
```

-
- Answer: Customer support issues peaked in July (Month 7) and June (Month 6), with 6 issues each, suggesting a mid-year rise in customer problems. This pattern could indicate seasonal stress points or service failures during summer, and targeted support initiatives during these months could proactively reduce issue volume.

- I need a report showing aircraft types with the highest number of seat bookings.

```
-- 5. I need a report showing aircraft types with the highest number of seat bookings.
select
    a.Model,
    count(bs.SeatID) as TotalSeatBookings
from
    BookingSeat bs
join
    Seat s on bs.SeatID = s.SeatID
join
    Flight f on s.FlightID = f.FlightID
join
    Aircraft a on f.AircraftID = a.AircraftID
group by
    a.Model
order by
    TotalSeatBookings desc;
```

-

- Answer: The "Music Jet" model had the highest number of seat bookings (6), followed by several other models like "Son Jet," "Skill Jet," and "Operation Jet" with 4 bookings each. This suggests that the "Music Jet" is currently the most utilized aircraft model, and prioritizing its maintenance and fleet availability could help sustain high booking rates.
- I need a list of customers with the highest number of flights booked over the past year.

-- 6. I need a list of customers with the highest number of flights booked over the past year.

```
select
    c.CustomerID,
    count(fb.BookingID) as FlightsBooked
from
    Customer c
join
    FlightBooking fb on c.CustomerID = fb.CustomerID
where
    year(fb.BookingDate) = 2025
group by
    c.CustomerID
order by
    FlightsBooked desc
limit 10;
```

- Answer: Customer ID 32 had the highest number of flights booked (4 flights) in 2025, with several other customers close behind at 2–3 flights each. This suggests a small but valuable group of frequent flyers the loyalty program team could target for personalized rewards to encourage even more repeat bookings.

- I need to analyze the average sale price per seat class (Economy, Business, First).

-- 7. I need to analyze the average sale price per seat class (Economy, Business, First).

```
select
    SeatClass,
    avg(BasePrice) as AveragePrice
from
    Seat
group by
    SeatClass;
```

- Answer: Surprisingly, Economy class had the highest average seat price (\$511), while Business class was lower at \$352 and First class averaged \$453. This unusual pricing structure suggests a need to review seat pricing strategy, as customers may be paying premium rates for Economy without the corresponding upgrade in service value.
- I need to see the total number of checked baggage items per month.

```

-- 8. I need to see the total number of checked baggage items per month.
select
    Month(f.DepartureTime) as Month,
    count(b.BaggageID) as TotalBaggageItems
from
    baggage b
join
    Flight f on b.FlightID = f.FlightID
group by
    Month(f.DepartureTime)
order by
    Month;

```

-
- Answer: December (Month 12) showed the highest volume of checked baggage items (12), while earlier months like January and February had much lower volumes (1–3 bags). This trend suggests a strong seasonal surge in baggage handling needs during the holiday period, signaling the need for additional staffing and logistics planning for end-of-year travel.

- I need a list of top hotels by number of bookings made through our platform.

```

-- 9. I need a list of top hotels by number of bookings made through our platform.
select
    Hotel_ID,
    count(*) as TotalBookings
from
    Hotel_booking
group by
    Hotel_ID
order by
    TotalBookings DESC
limit 10;

```

-
- Answer: The booking data shows that all top 10 hotels each had exactly one booking, meaning no hotel currently stands out as a dominant performer.

This indicates a lack of booking volume, suggesting the platform could focus marketing efforts on promoting preferred hotel partners to drive more bookings.

- I need a report of all flight bookings with a 'Pending' payment status to ensure timely collections.

```
-- 10. I need a report of all flight bookings with a 'Pending' payment status to ensure timely collections.
select
    BookingID,
    CustomerID,
    FlightID,
    BookingDate
from
    FlightBooking
where
    PaymentStatus = 'Pending';
```

-
- Answer: There are a significant number of flight bookings across multiple customers that remain unpaid. This trend suggests an urgent need for the finance or customer service team to implement payment recovery efforts, especially focusing on older pending accounts before they become unlikely to close.

Conclusion

The Expedia Travel Booking Database was designed to centralize and streamline the management of customer, flight, loyalty, booking, and payment data. By building a scalable, normalized relational database, Expedia can ensure faster access to key business information, improved accuracy of customer interactions, and better reporting for operational and strategic decision-making.

This database empowers stakeholders across the company to generate insights that drive customer retention, optimize flight and hotel offerings, and increase profitability. With strong entity relationships, clear business rules, and robust query capabilities, the system supports both day-to-day operations and long-term strategic planning, helping Expedia maintain its competitive advantage in the global travel marketplace.